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Unit 4 - Dive into Flow

PDF Documents

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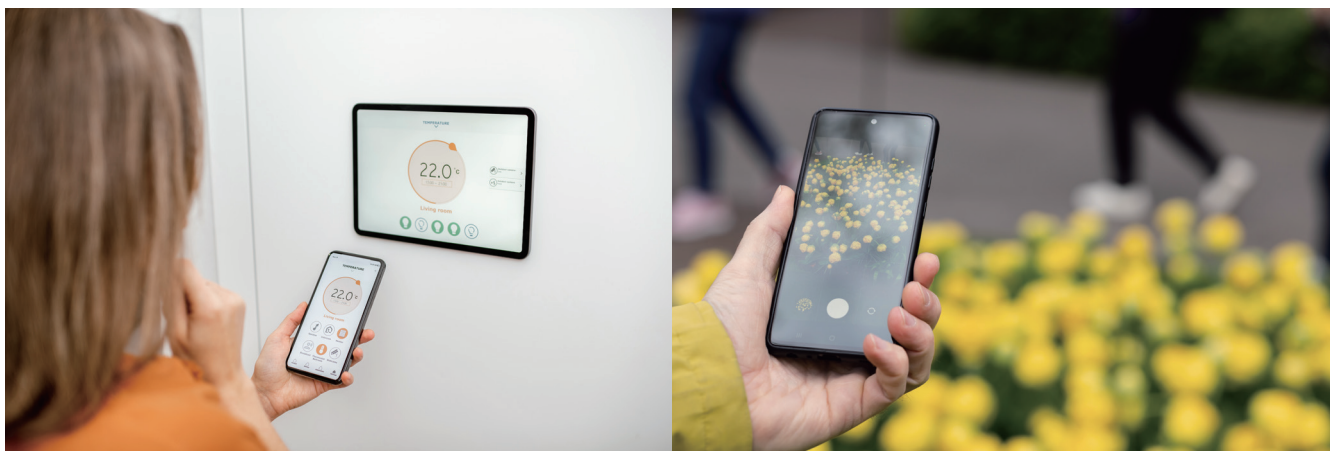


UNIT 4 DIVE INTO FLOW

SENSORS IN EVERYDAY LIFE

Sensors are all around us, helping to make our lives easier and safer. They seem like magical devices, but they're based on simple ideas that help us understand the world better. This section will explore how sensors are used daily and why they are essential.

Sensors in Action:



AUTOMATIC LIGHTS

How They Work: Motion sensors detect when someone enters a room. When movement is detected, the sensor sends a signal to turn on the light.

Why It's Useful: This saves energy because the lights only turn on when someone is in the room and turn off when no one is there.

THERMOSTATS

How They Work: Temperature sensors measure the room's temperature. The thermostat then adjusts the heating or cooling system to keep the room at the right temperature.

Why It's Useful: It helps keep our homes comfortable by ensuring the temperature is right.

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SMARTPHONES

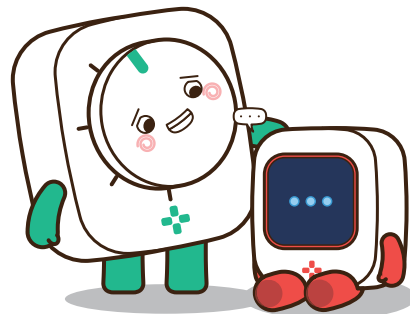
How They Work: Smartphones have several sensors, such as accelerometers that detect movement and proximity sensors that detect when you're holding the phone close to your ear.

Why It's Useful: These sensors help the phone adjust its screen brightness or turn off the display during calls to save battery.

CARS

How They Work: Modern cars have sensors that help with parking, like backup sensors that beep when you're getting close to something behind you.

Why It's Useful: They help drivers park safely and avoid collisions.





Multiple Choice Questions

1 What is the main purpose of a car backup sensor?

- ☐ A To measure the speed of the car
- ☐ B To detect obstacles behind the car
- ☐ C To check the fuel level
- ☐ D To control the car's air conditioning

CORRECT ANSWER

☒ B

2 Which module can sense the distance?

- ☐ A ToF
- ☐ B Button
- ☐ C IMU
- ☐ D Speaker

CORRECT ANSWER

☒ A

3 What is a common feature of backup sensors when they detect an obstacle?

- ☐ A The car's engine automatically turns off.
- ☐ B A warning sound or alert is activated.
- ☐ C The car's headlights turn on.
- ☐ D The car's radio volume increases.

CORRECT ANSWER

☒ B

Reflection Questions

1 How do sensors help us in our everyday lives?

2 Can you think of other devices at home or in public that use sensors? How do sensors help us in our everyday lives?

3 Can you think of other devices at home or in public that use sensors?